King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 102 Major Exam I Term 252

Original exam date: 17 February 2026. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. The area of the region bounded by the graphs of the equations $x = y^2$ and $x = y + 2$ is

(a) $\dfrac{11}{2}$

(b) $\dfrac{3}{2}$

(c) $\dfrac{5}{2}$

(d) $\dfrac{7}{2}$

(e) $\dfrac{9}{2}$

2. The area of the region bounded by the graphs of $f(x) = \dfrac{-4}{x^3}$, $y = 0$, $x = -3$, and $x = -1$ is

(a) $\dfrac{16}{9}$

(b) $\dfrac{9}{25}$

(c) $\dfrac{9}{16}$

(d) $\dfrac{25}{9}$

(e) $\dfrac{9}{4}$

3. $\int_{1/2}^{1} \operatorname{sech}^2(2x-1)\,dx =$

(a) $\dfrac{1}{2}\tanh 1$

(b) $\dfrac{1}{2}\coth 1$

(c) $\tanh 1$

(d) $\dfrac{1}{2}\operatorname{sech} 1$

(e) $\operatorname{sech} 1$

4. The lower sum (in terms of n, which is the number of equal subintervals), for the region bounded by the graph of the function $f(x) = 9 - x^2$ and the x-axis on the interval $[0, 2]$ is:

(a) $\dfrac{46}{3} - \dfrac{4}{n} + \dfrac{2}{3n^2}$

(b) $\dfrac{46}{3} + \dfrac{4}{n} - \dfrac{4}{3n^2}$

(c) $\dfrac{23}{3} + \dfrac{4}{n} + \dfrac{2}{3n^2}$

(d) $\dfrac{46}{3} + \dfrac{4}{n} + \dfrac{2}{3n^2}$

(e) $\dfrac{46}{3} - \dfrac{4}{n} - \dfrac{4}{3n^2}$

5. The average value of the function $f(x) = \sin x$ over the interval $[0, \pi]$ is

(a) $\dfrac{1}{\pi}$

(b) $-\dfrac{2}{\pi}$

(c) 0

(d) $\dfrac{2}{\pi}$

(e) $-\dfrac{1}{\pi}$

6. $\int x\sqrt{x+6}\,dx =$

(a) $\dfrac{2}{5}(x+6)^{1/2}(x-4) + C$

(b) $\dfrac{2}{5}(x+6)^{3/2}(x-4) + C$

(c) $\dfrac{2}{5}(x+6)^{3/2}(x+4) + C$

(d) $\dfrac{1}{5}(x+6)^{3/2}(x-4) + C$

(e) $\dfrac{1}{5}(x+6)^{3/2}(x+4) + C$

7. Possible values of a and b such that $\int_a^b \sin x\,dx < 0$ are:

(a) $a = \dfrac{\pi}{2},\ b = \pi$

(b) $a = 0,\ b = \dfrac{3\pi}{2}$

(c) $a = \dfrac{\pi}{2},\ b = 2\pi$

(d) $a = \dfrac{\pi}{2},\ b = \dfrac{3\pi}{2}$

(e) $a = 0,\ b = \dfrac{\pi}{2}$

8. If the interval $[0, 8]$ is partitioned at $x_0 = 0$, $x_1 = 1$, $x_2 = 3$, $x_3 = 7$ and $x_4 = 8$ with $c_1 = 1$, $c_2 = 2$, $c_3 = 5$ and $c_4 = 8$ (c_i is the sample point in the ith subinterval), then the Riemann sum of $f(x) = x^2 + 3x$ on $[0, 8]$ is equal to

(a) 270

(b) 274

(c) 272

(d) 278

(e) 276

9. Use the right end points and 3 rectangles to find an approximation of the area of the region between the graph of the function $f(x) = 2x^2 - x - 1$ and the x-axis over the interval $[2, 5]$.

(a) 87

(b) 85

(c) 84

(d) 86

(e) 88

10. $\displaystyle\int \frac{1}{\sqrt{-x^2 - 4x}}\,dx =$

(a) $\arcsin\left(\dfrac{x+4}{2}\right) + C$

(b) $\arccos\left(\dfrac{x+2}{2}\right) + C$

(c) $\arcsin(x+2) + C$

(d) $\arcsin\left(\dfrac{x+2}{2}\right) + C$

(e) $\arccos\left(\dfrac{x+4}{2}\right) + C$

11. $\int_0^5 |2x-5|\,dx =$

(a) $\dfrac{19}{2}$

(b) $\dfrac{27}{2}$

(c) $\dfrac{21}{2}$

(d) $\dfrac{23}{2}$

(e) $\dfrac{25}{2}$

12. If $F(x) = \int_0^{\sin x} \sqrt{t}\,dt$, then $F'\left(\dfrac{\pi}{6}\right) =$

(a) $\dfrac{\sqrt{2}}{2}$

(b) $\dfrac{\sqrt{3}}{2}$

(c) $\dfrac{\sqrt{6}}{4}$

(d) $\dfrac{\sqrt{6}}{2}$

(e) $\dfrac{\sqrt{6}}{3}$

13. $\int_{\pi/8}^{\pi/4} (\csc 2\theta - \cot 2\theta)\,d\theta =$

(a) $\dfrac{1}{2}\ln\left(1+\dfrac{\sqrt{2}}{4}\right)$

(b) $\dfrac{1}{2}\ln\left(1+\sqrt{2}\right)$

(c) $\dfrac{1}{2}\ln\left(1+\dfrac{\sqrt{2}}{2}\right)$

(d) $\ln(1+\sqrt{2})$

(e) $\ln\left(1+\dfrac{\sqrt{2}}{2}\right)$

14. $\int \dfrac{x^2 - 3x + 2}{x + 1}\, dx =$

(a) $\dfrac{x^2}{2} + 4x - 6\ln|x+1| + C$

(b) $\dfrac{x^2}{2} - 4x - 3\ln|x+1| + C$

(c) $\dfrac{x^2}{2} - 4x + 6\ln|x+1| + C$

(d) $x^2 - 2x + 6\ln|x+1| + C$

(e) $\dfrac{x^2}{2} + 4x + 3\ln|x+1| + C$

Answer Key

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Q	Ans	Brief explanation
1	**(e)**	Intersect at $y=-1,2$: $\int_{-1}^{2}(y+2-y^2)\,dy=\frac{9}{2}$.
2	**(a)**	$f>0$ on $[-3,-1]$: $\left[2x^{-2}\right]_{-3}^{-1}=2-\frac{2}{9}=\frac{16}{9}$.
3	**(a)**	$u=2x-1$: $\frac{1}{2}\int_0^1 \operatorname{sech}^2 u\,du=\frac{1}{2}\tanh 1$.
4	**(e)**	Decreasing, so right end points: $18-\frac{8}{n^3}\sum i^2=\frac{46}{3}-\frac{4}{n}-\frac{4}{3n^2}$.
5	**(d)**	$\frac{1}{\pi}\int_0^{\pi}\sin x\,dx=\frac{2}{\pi}$.
6	**(b)**	$u=x+6$: $\frac{2}{5}u^{5/2}-4u^{3/2}=\frac{2}{5}(x+6)^{3/2}(x-4)+C$.
7	**(c)**	$\int_a^b \sin x\,dx=\cos a-\cos b$; only $[\pi/2,2\pi]$ gives $-1<0$.
8	**(c)**	Widths $1,2,4,1$; $4+20+160+88=272$.
9	**(b)**	$\Delta x=1$, right end points $3,4,5$: $14+27+44=85$.
10	**(d)**	$-x^2-4x=4-(x+2)^2$, so the integral is $\arcsin\frac{x+2}{2}+C$.
11	**(e)**	Split at $x=\frac{5}{2}$: $\frac{25}{4}+\frac{25}{4}=\frac{25}{2}$.
12	**(c)**	$F'(x)=\sqrt{\sin x}\cos x$, so $F'(\pi/6)=\frac{1}{\sqrt{2}}\cdot\frac{\sqrt{3}}{2}=\frac{\sqrt{6}}{4}$.
13	**(c)**	$u=2\theta$: $\frac{1}{2}\left[\ln\lvert\csc u-\cot u\rvert-\ln\sin u\right]_{\pi/4}^{\pi/2}=\frac{1}{2}\ln\left(1+\frac{\sqrt{2}}{2}\right)$.
14	**(c)**	$\frac{x^2-3x+2}{x+1}=x-4+\frac{6}{x+1}$, giving $\frac{x^2}{2}-4x+6\ln\lvert x+1\rvert+C$.